

## ← Project review - ML1\_Introduction

|                                                                                                       |            |
|-------------------------------------------------------------------------------------------------------|------------|
|  Type of project     | Individual |
|  Duration            | 30 min     |
|  Passed Peer Reviews | 0/3        |

### Git project



### Student



lynnehea

level 5

### About

#### Introduction

School 21 methodology only makes sense if peer-to-peer reviews are taken seriously. Please read all the guidelines carefully before starting the review.

- Please remain courteous, polite, respectful, and constructive in all communications during this review.

- Point out possible flaws in the person's work and take the time to discuss and debate them.
- Keep in mind that sometimes there may be differences in interpretation of tasks and scope of functionality. Please remain open to each other's vision.

### Guidelines

- Evaluate only the files located in the src folder of the student's or group's GIT repository.
- If you have not finished the project yet, it is compulsory to read the entire instruction before starting the review.
- Ensure to start reviewing a group project only when the team is present in full.
- Use special flags in the checklist to report, for example, "empty work" if the repository does not contain the student's (or group's) work in the src folder of the develop branch, or "cheat" in the case of cheating, or if the student (or group) is unable to explain their work at any time during the review, or if any of the items below are not met. However, except in cases of cheating, you are encouraged to continue reviewing the project to identify the problems that caused the situation so that they can be avoided in the next review.
- Double check that the GIT repository is the one corresponding to the student or group.
- Carefully check that nothing malicious has been used to fool you.
- In controversial cases, remember that the checklist determines only the general order of the check. The final decision on project evaluation remains with the reviewer.

## Main part

### Main part. Task 1

- 1.a. Verification examples
- 1.b. Supervised: Classification: 2, 4 (as multiclass with target = ), 5 (target = whether customer participates in promotion or not or whether customer buys products from promotion), 6 (target = defect or not), 8 (target = whether user opens link, multiclass is bad option here because number of pages is too big), 10 (if we have detected facts of malfunctions).

Supervised: Regression: 1, 3 (predict number of hours from last medica

Unsupervised: Clustering: 5 (task is to find users that will be in the

Unsupervised: Association: 5 (task is to find users who are similar to

Unsupervised: Dimensionality Reduction: 5 and 9 (we can use it together

This division is only an example and does not cover all possible cases

- 1.c. Multiclass classification is a classification task with more than two classes; for example, classifying a set of images of fruits that can be oranges, apples, or pears. Multiclass classification assumes that each sample is assigned to one and only one label: a fruit can be either an apple or a pear, but not both at the same time.

Multilabel classification assigns a set of target labels to each sample. This can be thought of as predicting properties of a data point that are not mutually exclusive, such as topics that are relevant to a document. A text might be about religion, politics, finance, or education, all at the same time, or none of them. Taken from <http://scikit-learn.org/stable/modules/multiclass.html>.

- 1.d. Theoretical example is a regression problem. We can reduce a regression problem to a classification problem by stratifying our target. In other words, it is possible to divide a continuous target into discrete groups.



### Main part. Task 2

- 2.c. (49352, 15)
- 2.d. Target is Price.
- 2.e. There are no empty columns.



### Main part. Task 3

- 3.b.i. All values are in 0? There are some large values. We need to find outliers. (materials/etalons/3bi.png).
- 3.b.ii. Boxplot shows there are outliers (materials/etalons/3bii.png).
- 3.b.iv. Removing the outliers allows to see a good distribution. You need to remove outliers when working with linear regression (materials/etalons/3biv.png).
- 3.c.i. Object type.

- 3.c.ii. Three values — it is necessary to decode: low 33697 medium 11116 high 3566
- 3.c.iv. There are no outliers for bathrooms and bedrooms (materials/etalons/3civ-1.png) (materials/etalons/3civ-2.png).
- 3.d.i. The maximum correlation with bathrooms (materials/etalons/3di.png).
- 3.d.ii (materials/etalons/3dii-1.png)(materials/etalons/3dii-2.png).



#### Main part. Task 4

- 4.a There are no greater correlations with squared features (materials/etalons/4a.png).



#### Main part. Task 5

- 5.e. Peer's metrics may be some differ. It depends on sklearn version. Below you see approximately right metrics (materials/etalons/5e.png).



#### Fails

#### Record of the online-review

Select a file or drag it here

Allowed file types are: .avi, .mov, .mp4, .webm and file size: max 800 MiB

### Comment

Leave a comment...

✓ Review